

Ridge Test Procedure

## Ground Station

### Software

1. Boot up GS raspberry pi and make sure IP address is configured for a personal hotspot (so it can be run outside)
   a. Connect GS and laptop to same hotspot
   b. Use HDMI for GS (connect to monitor and change wifi network, record IP address)
2. Open VSCode SSH host
3. Go to /documents
4. Open terminal, enter cd GSW-Backend
5. Run python main.py
   a. MAKE SURE FREQUENCY IS RIGHT (433 vs 435)
6. Enter enter (or r enter r enter) for normal operation
   a. Typically, we want to look at the RSSI value to see signal strength

### Hardware

1. Plug GS into portable battery
2. Turn on GS using the power switch on the black box
3. Make sure antenna/attenuators are connected properly
   a. Most common: eggbeater antenna connect via coaxial cable with attenuation connected between the coaxial cable and the GS
4. If transmitting and receiving (ping-ponging), GS will make clicking sounds to signal transmit/receive at regular intervals

## CubeSat

1. Make sure CubeSat is connected to laptop
2. Connect to the mainboard by opening terminal and running “sudo minicom -D /dev/tty.usbmodem101 -b 115200” (can tab to autocomplete)
3. CTRL-D to reload